

Binary LP

$$\min c^T x$$

$$\text{s.t. } Ax = b$$

$$x \geq 0$$

$$x \in \{0, 1\}^q$$

EX Vertex Cover

...

$$\text{s.t. } x_i + x_j \geq 1$$

...

$$x \in \{0, 1\}^q$$

Having an “And” variable

$$x \wedge y \rightarrow z (= xy) (= x \wedge y)$$

$$x, y \geq z \geq x + y - 1$$

This allows for $z = 1$ iff x and y are 1 and 0 otherwise. This creates the flexibility for this constraint to be true without forcing $x, y = 1$

Misc

$$Q \in R^{q \times q}, x \in R^p$$

$$x^T Q x = \sum_i^p \sum_j^q q_{ij} x_i x_j$$

This is an arbitrary homogenous quadratic form.

EX

$$Q = [(1, 2), (3, 4)] \rightarrow x^T Q x = x_1^2 + 2x_1x_2 + 3x_1x_2 + 4x_2^2$$

We can assume Q is symmetric because it does not change the $x^T Q x$